Renal Failure and Renal Transplantation

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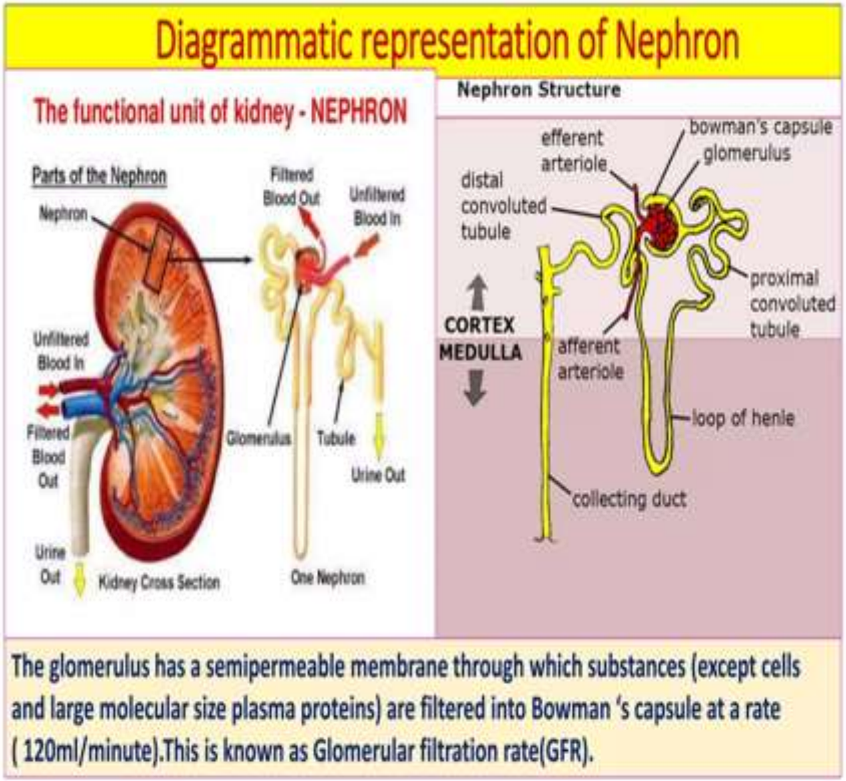
ILOs

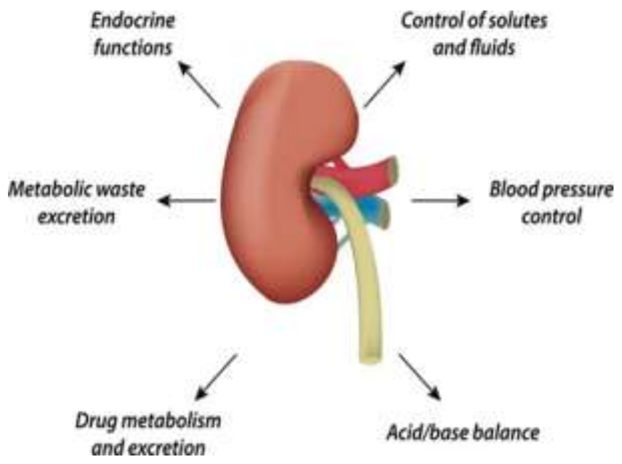
- **By the end of this presentation you should be able to:**
  - **Define Renal Failure (RF)**
  - **Classify RF (Acute, chronic RF)**
  - **List etiology (causes) and risk factors of RF.**
  - **Describe the clinical picture of RF.**
  - **Explain investigations available to reach diagnosis**
  - **Outline management of RF**
  - **Identify Dental Aspect of RF and Renal Transplantation**

Introduction

- **Nephron is the functional unit of the kidney (each kidney is composed of one million nephrons)**
- **Nephron is made of renal corpuscle (glomerulus and Bowman's capsule) and renal tubules**

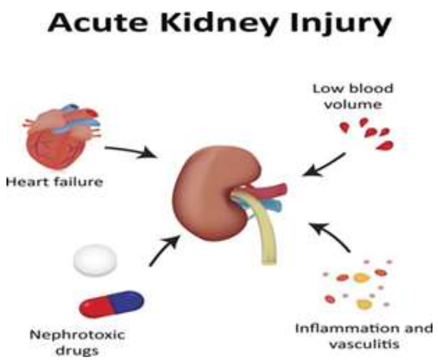


- **The kidneys (nephrons) have many functions:**
  - **Excretion of waste products.**
  - **Regulate fluid and electrolyte balance.**
  - **Regulate blood PH (Acid–base balance).**
  - **Endocrine function:**
    - **Renin production**
    - **Activation of Vitamin D.**
    - **Secretes hormones as erythropoietin**

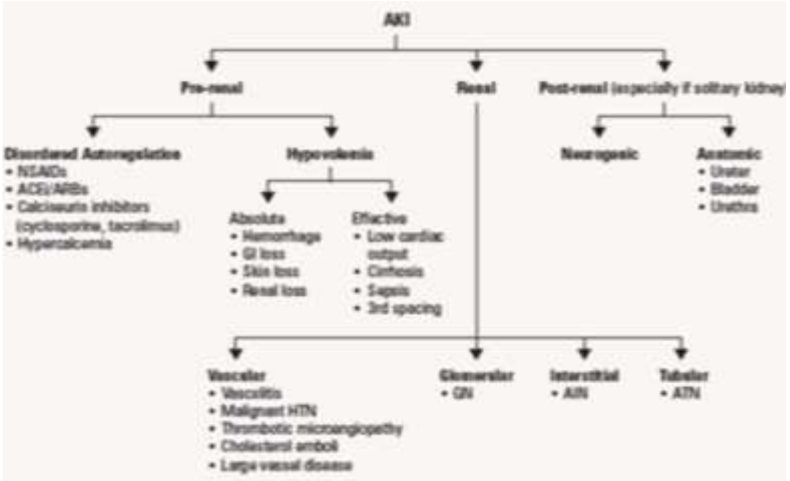


Acute kidney injury

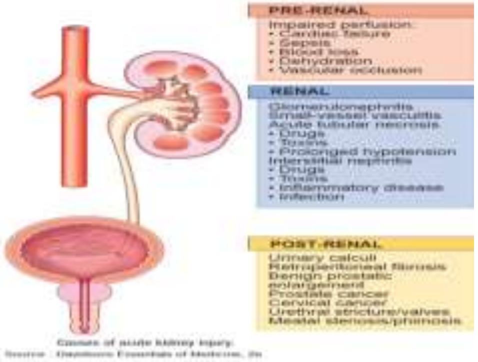
- **Abrupt deterioration or loss of renal function (within 48-72 hours) leading to retention of nitrogenous waste products, electrolyte disturbances.**
- **Usually reversible over days or weeks.**
- **AKI is a medical emergency, which may lead to disturbed conscious level (confusion, seizures and coma).**
- **Change in kidney function detected by S. creatinine.**



Causes of AKI



- **Pre-renal:**
  - **Reduced renal blood flow.**
- **Renal:**
  - **Parenchymal or vessels disorders.**
    - **Causes:**
      - **Inflammation, infections, toxins, drugs.**
- **Post renal:**
  - **Urinary tract obstruction (renal stone, Ureteric or urethral stricture, Prostatic hypertrophy).**
- **Management:**
  - **Treat underlying cause.**
  - **Dialysis if needed.**



Clinical features of AKI

- **Symptoms and Signs of underlying cause:**
  - **Prerenal causes:**
    - **Dry mouth, increased thirst, Pale, cold extremities, Slow capillary refill., Reduced and dark urine, Drowsiness and confusion.**
    - **Increased sympathetic tone: Tachycardia, Tachypnea, Weak rapid pulse, hypotensive, Sweating, low jugular venous pressure (except CHF),**

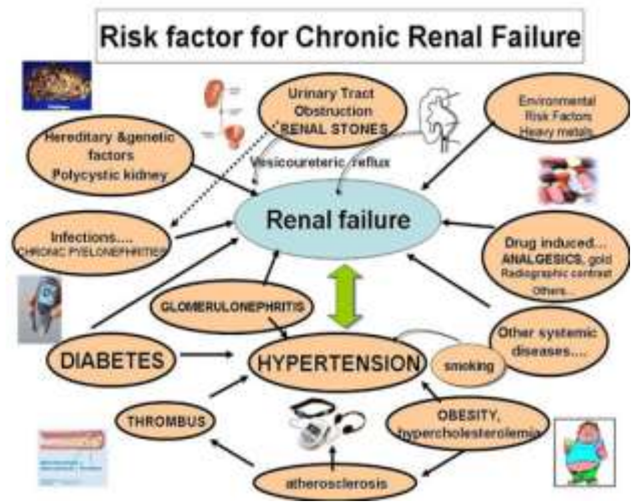
- **Symptoms of uremia:**
    - **Oliguria and hypertension.**
    - **Anorexia, nausea and vomiting**
    - **Breath odor (acidotic breathing) and metallic taste.**
    - **Weakness and arrhythmia due to hyperkalemia.**
    - **Uremic encephalopathy (disturbed conscious level).**
    - **Bleeding: Epistaxis and GIT hemorrhage.**

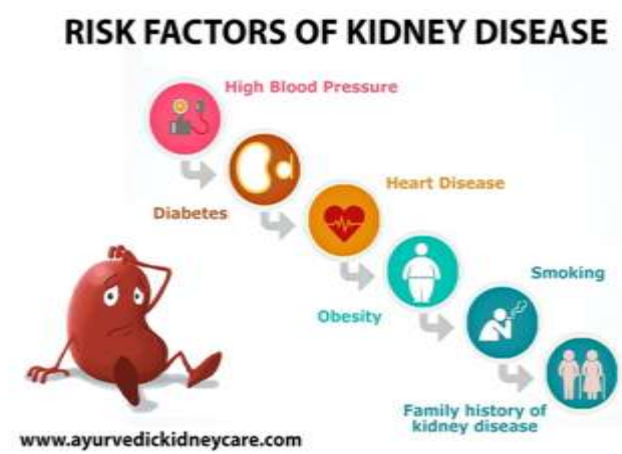
Chronic Kidney Disease

- **Chronic kidney disease (CKD): longstanding (>3 months) progressive and irreversible deterioration of kidney function.**
- **Change in kidney function detected by high level of S. creatinine and lower GFR (<60 ml/min)**
- **End stage renal disease (ESRD): represents a decline in kidney function requiring renal replacement therapy (RRT) which can occur over AKI, gradually (CKD), or acute on top of chronic.**

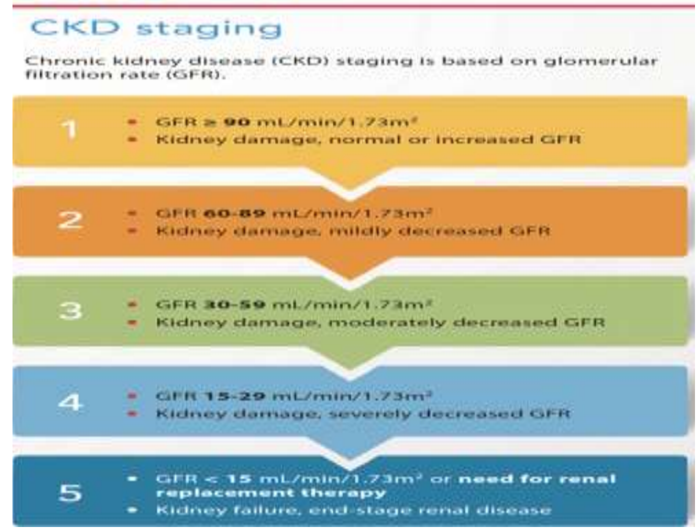
Causes of ESRD

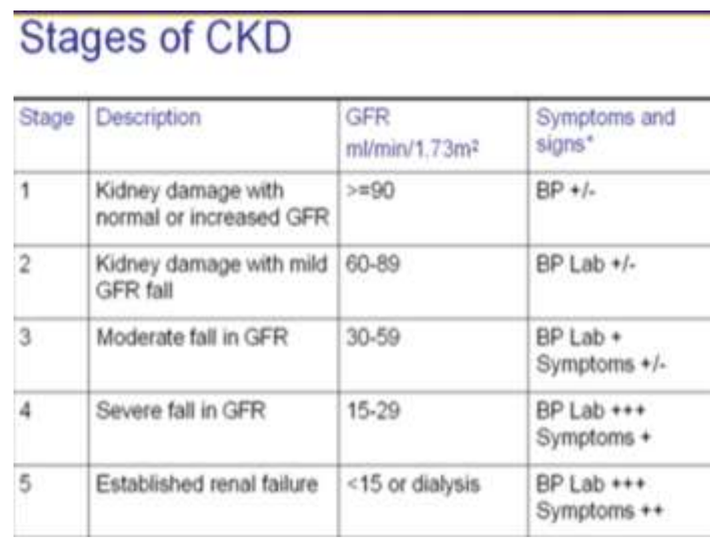
- **DM type 2 (42%), 4% type 1.**
- **Hypertension (28%).**
- **Glomerular disease (7%):**
  - **Primary: Focal glomerulosclerosis.**
  - **Secondary to systemic lupus erythematosus (SLE), thrombotic thrombocytopenic purpura (TTP), hemolytic uremic syndrome (HUS).**





- **Congenital (3%):**
  - **Poly cystic kidney disease**
- **Vascular disease:**
  - **Reno-vascular disease, vasculitis.**
- **Tubulo-interstitial nephritis (3%):**
  - **Interstitial nephritis, Pyelonephritis**
- **Urinary tract obstruction.**





Stages of CKD

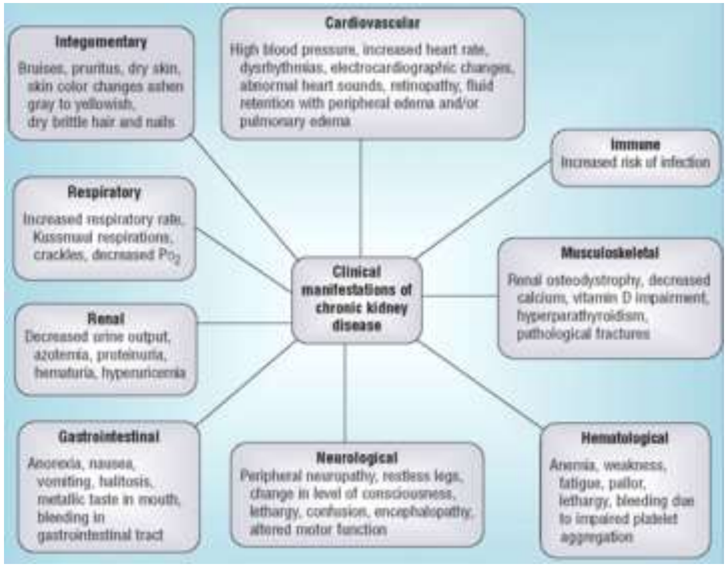
Stage	Description	GFR ml/min/1.73m²	Symptoms and signs*
1	Kidney damage with normal or increased GFR	>=90	BP +/-
2	Kidney damage with mild GFR fall	60-89	BP Lab +/-
3	Moderate fall in GFR	30-59	BP Lab + Symptoms +/-
4	Severe fall in GFR	15-29	BP Lab +++ Symptoms +
5	Established renal failure	<15 or dialysis	BP Lab +++ Symptoms ++

Clinical pictures of ESRD

- **Early: asymptomatic**
- **Late:**
  - **General:**
    - **Malaise, fatigue, loss of concentration.**
    - **Nocturia and polyuria due to impaired concentrating ability.**
    - **Later on, kidney lose loses of power to dilute urine.**
    - **Oliguria as CKD worse.**
    - **Signs of dehydration as dry tongue.**
    - **Acidotic breathing.**

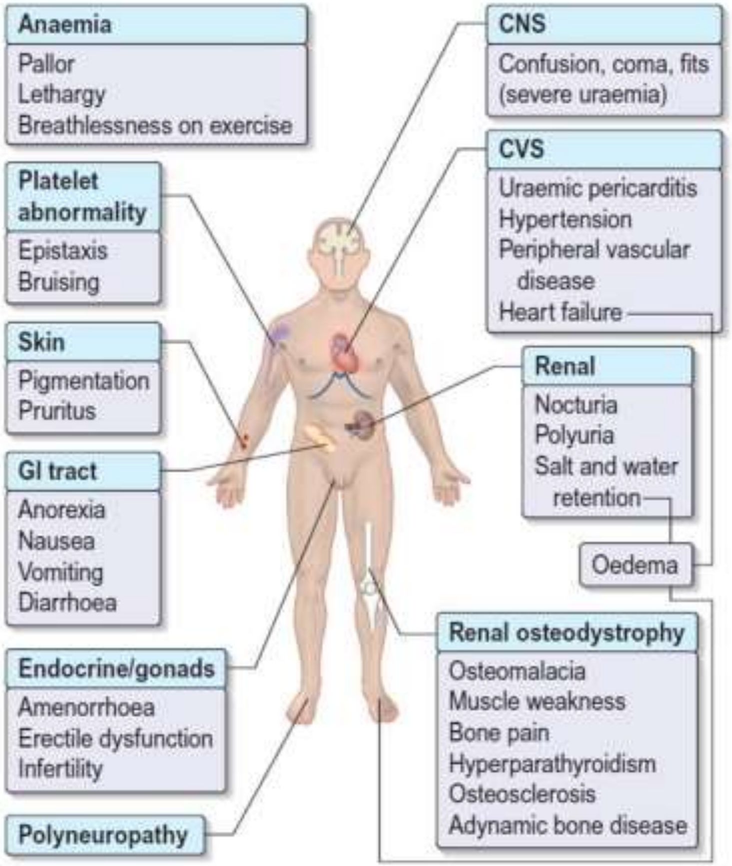
Anemia due to:

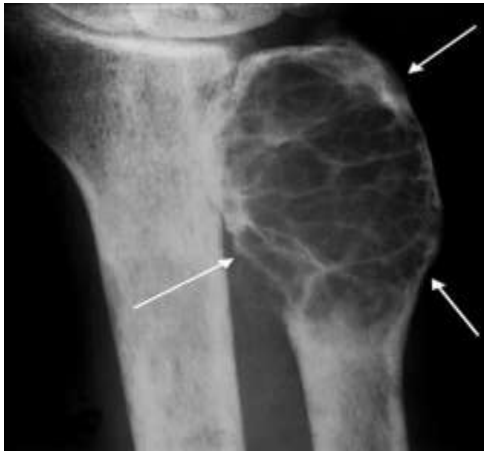
- **Erythropoietin deficiency**
- **Bone marrow depression from toxins & fibrosis from hyperparathyroidism.**
- **Increase blood loss due to:**
  - **GIT bleeding, hemodialysis, platelet dysfunction (coagulopathy).**



Renal osteodystrophy:

- **Phosphate retention due to Kidney disease leads to decrease plasma calcium and subsequently raised parathyroid hormone (secondary hyperparathyroidism).**
- **Also deficiency of renal production of 1,25-dihydroxycholecalciferol (vitamin D3)**





Oral manifestation of CKD

- **Dry mouth (xerostomia): reduce salivary flow lead to infection (Candidiasis).**
  - **Halitosis and a metallic taste.**
  - **Increase tooth decay.**
    - **To increase saliva, try chewing sugarless gum or sucking on sugarless candy.**

- **Uremic stomatitis**
- **Periodontal disease: Gingivitis, periodontitis**
- **Gingival overgrowth: due to side effect of some drugs as nifedipine and cyclosporine.**

- Petechiae and ecchymosis of oral mucosa and oral bleeding.

- Pallor of oral mucosa secondary to anemia
- Pigmentation of oral cavity

- Enamel hypoplasia

- Disturbed calcium and phosphate metabolism may cause enamel opacities, loss of lamina dura, loosening of teeth, bony fractures.
- Osteodystrophy (radiolucent jaw lesion)

Management

- **Clinical (history and examinations).**
- **Kidney function test: Increase plasma urea, and creatinine levels.**
- **Decrease GFR.**
- **CBC: Normocytic normochromic anemia, Thrombocytopenia**
- **High ESR: myeloma & vasculitis.**
- **Blood glucose level, Hb A1C**
- **S. electrolytes:**
    - **High ƒK+, PO3- ƒ - Low Na+, HCO3.**
    - **Low Ca2+ (↓Vit. D activation, ↑phosphate, secondary hyperparathyroidism, ↓Albumin). ƒ**
- **Arterial blood gas (ABG): Metabolic acidosis**

- **Urinalysis:**
  - **Hematuria, Proteinuria, RBCs, Granular casts, Red-cell casts**
  - **Urine electrolyte**
  - **GFR >90 ml/min/1.73m2 (normal)**
- **Radiological investigation**
    - **Renal Ultrasound, CT, MRI**
- **Renal biopsy**

Treatment

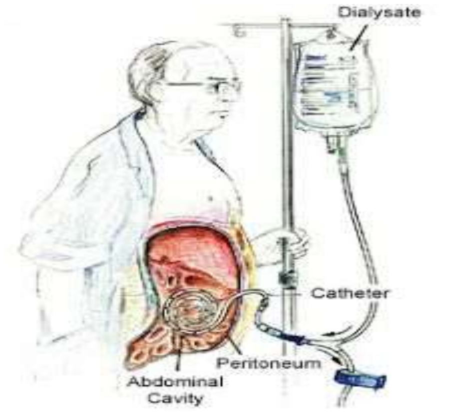
- **Aim: decrease progression of CKD and complication (cardiovascular risk, which is the main cause of mortality).**
- **Treat the underlying cause:**
    - **Blood pressure (BP) control.**
    - **Diabetes control: insulin or oral hypoglycemic drugs**
    - **Treat infection, obstruction**
    - **Stop nephrotoxic drugs (NSAIDs)**

- **Life style modifications:**
    - **Diet: potassium or phosphorous restriction, salt or water.**
    - **Smoking cessation especially for cardiovascular risk.**

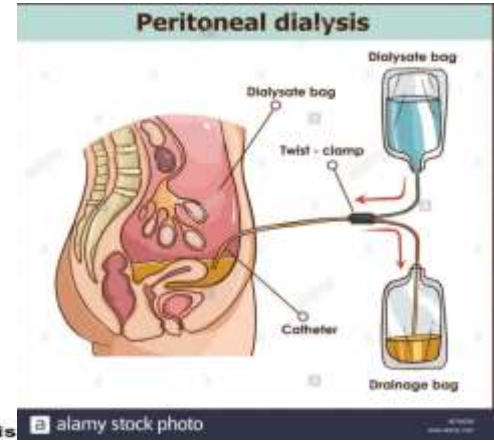
Treat symptoms and complications

- **Vomiting** **(metoclopramide)**
- **Metabolic acidosis:**
  - **Sodium bicarbonate.** *ƒ*
- **Calcium and phosphate disorders:** Š
  - **Calcium supplements (carbonate or acetate): binds phosphate in bowel and excreted.** Š
  - **Sevelamer (gut phosphate binder): reduce absorption of phosphate. Also attenuates vascular calcification and lower cholesterol levels.** Š
  - **Sensipar (Cinacalcet) (increase sensitivity of Ca. receptors in parathyroid to Ca2+, decreasing PTH)** *ƒ*
  - **Subtotal parathyroidectomy**

- **Anemia:**
  - **Erythropoietin stimulating agents (ESAs)**
  - **IV iron prior to ESAs and folic acid**
- **Renal Replacement Therapy**
  - **Renal dialysis (hemodialysis or peritoneal) or**
  - **Renal transplantation**



Continuous Ambulatory Peritoneal Dialysis



Renal transplantation

- Transplantation is now strongly recommended for all patients with ESRD (young and diabetic nephropathy patients).
- It enhance quality life and is more effective than chronic dialysis.
- Outcomes are improved by using a kidney from an unaffected relative (cadaveric or living donors) and performing the transplant before the patient needs to have dialysis.
- All recipients require lifelong immunosuppression (corticosteroid + steroid-sparing drug as azathioprine or cyclosporine) to prevent rejection.

- **Prior to a transplant, patients are generally screened and treated for all infections, including dental, to prevent post-transplant complications.**
- **Complications of transplantation includes:**
  - **Transplant rejection.**
  - **Increased risk of atheroma and ischemic heart disease**
  - **Ciclosporin-induced nephropathy**
  - **Immunosuppression induced infection or malignancy**
  - **New-onset DM or hypertension (due to prednisone use)**

DENTAL ASPECTS OF CKD

- Up to 90% patients with renal disease has oral symptoms.
- Keep good oral hygiene:
  - Brushing twice daily by a fluoridated toothpaste to strengthen teeth and prevent development of cavities. Also using antimicrobial rinses.
    - Brushing will help to remove excess plaque, as well as any acids that can harm tooth enamel.
  - Flossing once daily: prevent accumulation of food debris between the teeth and prevent the build up that can lead to tartar.
  - Visiting your dentist twice a year.
  - Full or partial dentures should be carefully cleaned daily and removed at night to prevent development of sore spots or ulcers.

- **Screening test (CBC, bleeding time)**
- **Monitor of blood pressure (avoid use of cuff in arm with shunt)**
- **Avoid dental treatment if disease unstable.**
- **Avoid dental treatment on day of dialysis (bleeding from heparin) and usually best carried out on the day after dialysis.**
  - **Hemostasis if surgical procedures are necessary.**
    - **Desmopressin (DDAVP) may provide hemostasis for up to 4 h.**
    - **If fails, cryoprecipitate, has a peak effect at 4-12 h and lasts up to 36 h.**

- **Avoid nephrotoxic drugs or reduce dose**
  - Nephrotoxic, especially in elderly patient or renal damage or cardiac failure.
- **Aspirin (75 mg given as prophylaxis against cardiovascular disease) and other NSAIDs should be avoided.**
  - Gastrointestinal irritation, peptic ulcer and bleeding associated with CKD.
  - Use of NSAIDs increase risk for acute renal failure, interstitial nephritis.
  - NSAIDs can cause peripheral edema, elevated blood pressure and exacerbation of heart failure.
- **Antihistamines may cause dry mouth or urinary retention.**

Nephrotoxic drugs

- Tetracyclines
- Streptomycin
- Vancomycin
- Gentamycin
- Acyclovir
- Acetaminophen
- Phenacetine
- NSAIDs
- Asprin
- Antihistamines,
- Phenobarbitones

Reduce dose

- Cephalosporins
- Penicillins
- Ampicillin
- Metronidazole
- Acyclovir
- Paracetamol
- Benzodiazpine

Usual dose

- Cloxacillin
- Erythromycin
- Minocycline
- Codiene
- Diazepam
- Lidocaine

- **Infections (should treated according to culture and sensitivity test) are poorly controlled by the patient with CKD, especially if immunosuppressed.**
  - **Tuberculosis is more frequent.**
  - **Hemodialysis predisposes to blood-borne viral infections such as hepatitis virus.**
  - **Odontogenic infections should be treated vigorously.**
    - **Erythromycin, flucloxacillin, and fucidin can be given in standard dosage.**
    - **Penicillin and metronidazole should be given in lower doses.**
    - **Tetracycline can worsen nitrogen retention and acidosis and are best avoided.**
      - **Doxycycline or minocycline may be given**

- **Systemic fluorides should not be given (fluoride excretion by damaged kidneys).**
    - **Fluorides can usually safely be used topically for caries prophylaxis.**
- **Antacids containing magnesium salts should not be given as there may be magnesium retention.**
    - **Antacids containing calcium or aluminium bases may impair absorption of penicillin V and sulfonamides.**
- **Major surgical procedures may be complicated by hyperkalemia (due to tissue damage, acidosis and blood transfusion) which predisposes to arrhythmias and may cause cardiac arrest.**

- **Local anesthesia is safe unless there is a severe bleeding tendency.**
- **Conscious sedation, relative analgesia.**
  - **Avoid intravenous medication in arm with shut because of the risk of consequent fistula infection or thrombophlebitis.**
  - **Never put a cannula into the arterio-venous fistula arm.**
  - **Midazolam is preferable because of the lower risk of thrombophlebitis.**

- **General anesthesia (myocardial depressant and hypotension) is contraindicated if the hemoglobin is below 10 g/dl.**
    - **Enflurane should be avoided if other nephrotoxic agents are used concurrently.**
    - **Isoflurane and sevoflurane are probably safer.**
    - **Thiopental followed by very light GA with nitrous oxide is the technique of choice.**

- **Osseous lesions include loss of the lamina dura, osteoporosis and osteolytic areas (renal osteodystrophy).**
- **In children with CRF:**
  - **Jaw growth is usually retarded and tooth eruption may be delayed.**
  - **Malocclusion and enamel hypoplasia with brownish discoloration, but tetracycline staining of the teeth should no longer be seen.**
  - **A lower caries rate and less periodontal disease in children with CKD.**
- **Mucosal lesions may be seen:**
  - **Oral mucosa may be pale (anemia) and may be oral ulceration.**

Dental Aspect of Renal Transplantation

- **Any oral sign of infection must be examined immediately and treated aggressively in an immunosuppressed transplant recipient.**
- **Hematological abnormalities, such as anemia and platelet dysfunction (bleeding).**
- **Viral hepatitis B and C.**
- **Cardiovascular disorders such as hypertension, atherosclerotic heart disease with myocardial infarction, congestive heart failure, and left ventricular hypertrophy.**
- **Erythromycin contraindicated in patient with cyclosporine.**
- **Cyclosporine cause gingival swelling.**

Thank you!!!